

Satellite Visibility Determination for a Train

Xiaojing Fan, Huachun Zhou, Tao Zheng, and Jingfu Yan *

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To determine the satellites visible to a train, the longitude and latitude information of LEO satellites first be calculated, followed by the visibility determination.

1.1 Calculation of Satellite Longitude and Latitude

Based on satellite Two-Line Element Set (TLE) data, the longitude and latitude, namely the subsatellite point coordinates, are calculated using orbital dynamics equations. A TLE data set contains orbital data of real LEO satellites currently in orbit and consists of two lines of text. The first line contains information such as the LEO satellite number, classification, epoch time, and the first and second derivatives of the mean motion. The second line contains the orbital inclination μ , right ascension of the ascending node (RAAN) η , eccentricity ζ , argument of perigee κ , and mean anomaly ϑ . The second line is mainly used to locate the LEO satellite. Since external perturbations cause the satellite orbit to drift continuously, the accuracy of TLE data decreases over time. These data are currently updated periodically and provided by Celestrak and Space-Track.

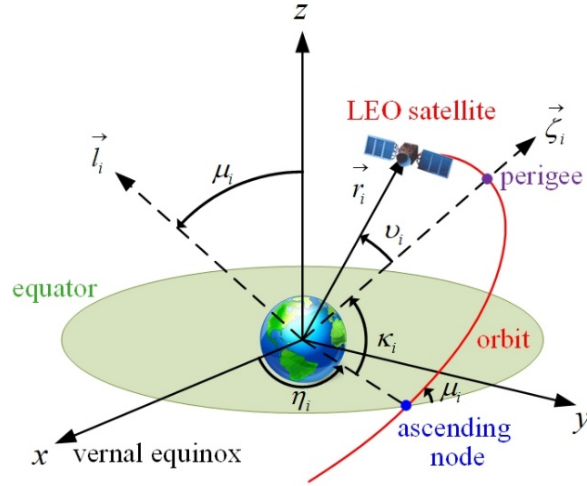


Figure 1: Schematic diagram of satellite orbital parameters.

As shown in Fig. 1, for satellite L_i , μ_i denotes the angle between the satellite orbital plane and the Earth's equatorial plane. $\mu_i \in [0^\circ, 180^\circ]$, where 0° represents an equatorial orbit and 90° represents a polar orbit. When $\mu_i \in [0^\circ, 90^\circ]$, the orbit is prograde and its direction is consistent with the Earth's rotation. When $\mu_i \in [90^\circ, 180^\circ]$, the orbit is retrograde and its direction is opposite to the Earth's rotation. The inclination significantly affects the coverage area of a LEO satellite. $\eta_i \in [0^\circ, 360^\circ]$ denotes the azimuth of the satellite orbit relative to the vernal

*Xiaojing Fan, Huachun Zhou, Tao Zheng, and Jingfu Yan are with the Institute of Electronic and Information Engineering, Beijing Jiaotong University, Beijing, 100044, China. E-mail: {23111043, hchzhou, zhengtao, 22110030}@bjtu.edu.cn

equinox direction, i.e., the angle at which the satellite crosses the equator from south to north at the ascending node. $\zeta_i \in [0, 1]$ describes the geometric shape of the satellite orbit around the Earth. The direction of the eccentricity vector $\vec{\zeta}_i$ is from the center of the orbit to the perigee, where the satellite is closest to the Earth. κ_i denotes the orientation of the satellite elliptical orbit in space and defines the angle from the ascending node to the perigee. ϑ_i describes the satellite's position along its orbital path at a specific time, which is called the epoch. It should be noted that the mean anomaly approximately represents the satellite's position in its orbit under the assumption that the satellite moves at a constant speed along a circular orbit. $\mathcal{M}_i$ is not equal to the true anomaly $\nu_i(t)$, which accurately reflects the satellite's actual position in the orbit. The mean anomaly is used to calculate the eccentric anomaly $\Omega_i(t)$, from which the true anomaly can be further obtained. $\vec{l}_i$ denotes the angular momentum vector, and $\vec{r}_i$ denotes the position vector of the satellite relative to the geocenter.

To determine the satellites visible to the train, the satellite's Earth-fixed coordinates and longitude/latitude at time t must first be calculated from its orbital parameters. Based on TLE orbital parameters, the satellite position is obtained through the transformation process from orbital-plane coordinates to the celestial inertial system and then to the Earth-fixed coordinate system. Since this process belongs to standard orbital-dynamics computation, its detailed derivation is given in Appendix A. Based on the Earth-fixed coordinates of the satellite and the train, the elevation angle can be further calculated to complete the visibility determination.

1.2 Visible Satellite Determination

Assume that the longitude and latitude of satellite L_i are $(L_i^\lambda(t), L_i^\phi(t))$, its Earth-fixed coordinate is $L_i(t)$, and the longitude and latitude of train T_j are $(T_j^\chi(t), T_j^\psi(t))$. We calculate the Earth-fixed coordinate $T_j(t)$ of the train. The train longitude and latitude are converted to radians as

$$\begin{aligned} t_j^\chi(t) &= T_j^\chi(t) \cdot \frac{\pi}{180}, \\ t_j^\psi(t) &= T_j^\psi(t) \cdot \frac{\pi}{180}. \end{aligned} \quad (1)$$

The prime vertical radius of curvature is calculated as

$$N_j(t) = \frac{a_E}{\sqrt{1 - e^2 \sin^2(t_j^\psi(t))}}. \quad (2)$$

According to the WGS-84 ellipsoid parameters, $a_E = 6378137$ m is the Earth's semi-major axis, e is the Earth's eccentricity, and

$$e^2 = 2f - f^2, \quad f = \frac{1}{298.257223563},$$

where f is the Earth's flattening.

The Earth-fixed coordinate of the train is

$$T_j(t) = \begin{bmatrix} x_{t_j}(t) \\ y_{t_j}(t) \\ z_{t_j}(t) \end{bmatrix}. \quad (3)$$

where

$$\begin{aligned} x_{t_j}(t) &= (N_j(t) + h_j) \cos(t_j^\psi(t)) \cos(t_j^\chi(t)), \\ y_{t_j}(t) &= (N_j(t) + h_j) \cos(t_j^\psi(t)) \sin(t_j^\chi(t)), \\ z_{t_j}(t) &= (N_j(t)(1 - e^2) + h_j) \sin(t_j^\psi(t)). \end{aligned} \quad (4)$$

Here, h_j is the train altitude.

For satellite L_i and train T_j , their difference vector is

$$\begin{aligned} x_{i,j}(t) &= x_{l_i}(t) - x_{t_j}(t), \\ y_{i,j}(t) &= y_{l_i}(t) - y_{t_j}(t), \\ z_{i,j}(t) &= z_{l_i}(t) - z_{t_j}(t). \end{aligned} \quad (5)$$

The Earth-fixed coordinate system is converted to the east-north-up (ENU) coordinate system as

$$\begin{aligned} E_{i,j}(t) &= -\sin(t_j^X(t))x_{i,j}(t) + \cos(t_j^X(t))y_{i,j}(t), \\ N_{i,j}(t) &= -\sin(t_j^\psi(t))\cos(t_j^X(t))x_{i,j}(t) - \sin(t_j^\psi(t))\sin(t_j^X(t))y_{i,j}(t) + \cos(t_j^\psi(t))z_{i,j}(t), \\ U_{i,j}(t) &= \cos(t_j^\psi(t))\cos(t_j^X(t))x_{i,j}(t) + \cos(t_j^\psi(t))\sin(t_j^X(t))y_{i,j}(t) + \sin(t_j^\psi(t))z_{i,j}(t). \end{aligned} \quad (6)$$

The elevation angle of satellite L_i relative to train T_j is

$$ele_{i,j}(t) = \frac{180}{\pi} \text{atan2}\left(U_{i,j}(t), \sqrt{(E_{i,j}(t))^2 + (N_{i,j}(t))^2}\right). \quad (7)$$

where atan2 is the four-quadrant inverse tangent function.

When

$$ele_{i,j} \geq ele_{\min},$$

the satellite is visible to the train.

Appendix A Satellite Position Calculation and Coordinate Transformation

This appendix presents the satellite position calculation process involved in the satellite visibility analysis in the main text. The related derivation is mainly based on classical orbital mechanics and coordinate transformation methods, and is used to calculate the Earth-fixed coordinates, longitude and latitude, and spatial position quantities required for subsequent visibility determination from TLE orbital parameters at any time. Since this part belongs to fundamental methodology, it is placed in the appendix to avoid affecting the compactness of the main text.

Appendix A.1 Mean Anomaly and Orbital Radius Calculation

Let the semi-major axis of the i -th satellite be a_i , and let its orbital period be δ_i . Then its mean motion is

$$\mathcal{H}_i = \sqrt{\frac{\varpi}{a_i^3}} = \frac{2\pi}{\delta_i}. \quad (A.1)$$

where ϖ is the Earth's standard gravitational parameter,

$$\varpi = GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2.$$

Here, $G = 6.673 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the universal gravitational constant, and $M_E = 5.974 \times 10^{24} \text{ kg}$ is the Earth's mass.

At any time t , the mean anomaly $\mathcal{M}_i$ of the satellite is

$$\mathcal{M}_i(t) = \mathcal{M}_{i0} + \mathcal{H}_i(t - t_{i0}). \quad (A.2)$$

Furthermore, from the first-order approximate solution of Kepler's equation, the eccentric anomaly Ω_i is obtained as

$$\Omega_i(t) = \mathcal{M}_i(t) + \zeta_i \sin \mathcal{M}_i(t). \quad (A.3)$$

where ζ_i is the eccentricity.

The true anomaly ν_i is

$$\nu_i(t) = 2 \tan^{-1} \left(\left[\frac{1 + \zeta_i}{1 - \zeta_i} \right]^{\frac{1}{2}} \tan \frac{\Omega_i(t)}{2} \right). \quad (\text{A.4})$$

The magnitude of the angular momentum is

$$\|\vec{l}_i\| = (\varpi a_i (1 - \zeta_i^2))^{\frac{1}{2}}. \quad (\text{A.5})$$

The distance between the satellite and the geocenter is

$$\|\vec{r}_i(t)\| = \frac{\|\vec{l}_i\|^2}{\varpi (1 + \zeta_i \cos(\nu_i(t)))}. \quad (\text{A.6})$$

Appendix A.2 Orbital-Plane Coordinate Representation

The position of the i -th satellite in the orbital-plane coordinate system, denoted by $\sigma_i(t)$, can be expressed as

$$\sigma_i(t) = \begin{bmatrix} \|\vec{r}_i(t)\| \cos(\nu_i(t)) \\ \|\vec{r}_i(t)\| \sin(\nu_i(t)) \\ 0 \end{bmatrix}. \quad (\text{A.7})$$

Appendix A.3 Coordinate Transformation from the Orbital Frame to the Earth-Fixed Frame

Through the RAAN η_i , inclination μ_i , and argument of perigee κ_i , three rotation matrices ρ_i^α , ρ_i^β , and ρ_i^γ are constructed to transform the satellite position from the orbital-plane coordinate system to the Earth-centered inertial coordinate system. Let the Earth's rotation angle be $\theta_i(t)$, obtained by integrating the Earth's angular rotation rate ω_E :

$$\theta_i(t) = \int_{t_0}^t \omega_E d\tau, \quad \omega_E = 7.292 \times 10^{-5} \text{ rad/s}.$$

Furthermore, the rotation matrix $\rho_i^\theta(t)$ is used to realize the transformation from the Earth-centered inertial coordinate system to the Earth-centered, Earth-fixed coordinate system. Therefore, the entire process can be expressed as

$$L_i(t) = \left[\rho_i^\alpha \rho_i^\beta \rho_i^\gamma \rho_i^\theta(t) \right] \sigma_i(t). \quad (\text{A.8})$$

Under the convention that column vectors are on the right and rotations follow a right-handed coordinate system, the rotation matrices are

$$\begin{aligned} \rho_i^\alpha &= \begin{bmatrix} \cos(\eta_i \frac{\pi}{180}) & -\sin(\eta_i \frac{\pi}{180}) & 0 \\ \sin(\eta_i \frac{\pi}{180}) & \cos(\eta_i \frac{\pi}{180}) & 0 \\ 0 & 0 & 1 \end{bmatrix}, & \rho_i^\beta &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\mu_i \frac{\pi}{180}) & -\sin(\mu_i \frac{\pi}{180}) \\ 0 & \sin(\mu_i \frac{\pi}{180}) & \cos(\mu_i \frac{\pi}{180}) \end{bmatrix}, \\ \rho_i^\gamma &= \begin{bmatrix} \cos(\kappa_i \frac{\pi}{180}) & -\sin(\kappa_i \frac{\pi}{180}) & 0 \\ \sin(\kappa_i \frac{\pi}{180}) & \cos(\kappa_i \frac{\pi}{180}) & 0 \\ 0 & 0 & 1 \end{bmatrix}, & \rho_i^\theta(t) &= \begin{bmatrix} \cos(\theta_i(t)) & -\sin(\theta_i(t)) & 0 \\ \sin(\theta_i(t)) & \cos(\theta_i(t)) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (\text{A.9})$$

Appendix A.4 Longitude and Latitude Calculation

The satellite Earth-fixed coordinate $L_i(t)$ is a 3×1 column vector. It is written as

$$L_i(t) = \begin{bmatrix} x_{l_i}(t) \\ y_{l_i}(t) \\ z_{l_i}(t) \end{bmatrix}. \quad (\text{A.10})$$

Then its longitude $L_i^\lambda(t)$ and latitude $L_i^\phi(t)$ are respectively

$$\begin{aligned} L_i^\lambda(t) &= \text{atan2}(y_{l_i}(t), x_{l_i}(t)), \\ L_i^\phi(t) &= \sin^{-1} \left(\frac{z_{l_i}(t)}{\|L_i(t)\|_{L_2}} \right). \end{aligned} \quad (\text{A.11})$$

where

$$\|L_i(t)\|_{L_2} = \left(\sum_{n=1}^3 |L_i(t)[n]|^2 \right)^{\frac{1}{2}}$$

is the L_2 norm, namely the Euclidean norm, of $L_i(t)$.